Spectrometer Live Tool – User Guide

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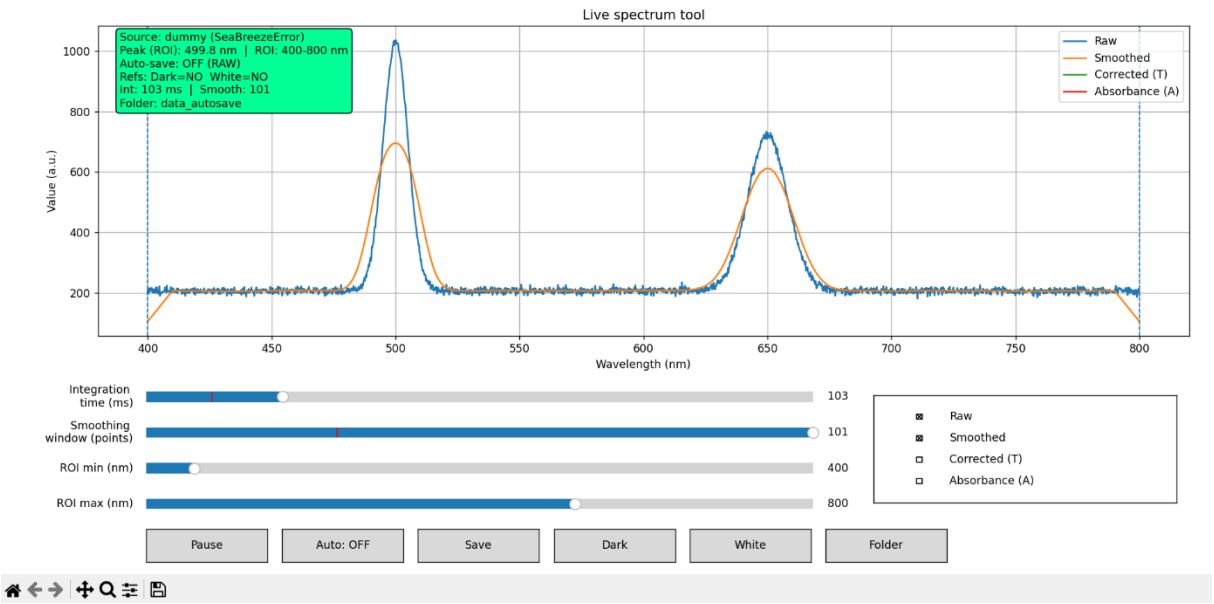
1. What is this tool?

This application is a live spectrometer interface designed for laboratory measurements and teaching demonstrations. It allows users to visualize spectra in real time, apply reference-based corrections, and export measurement data without modifying the source code.

2. How to start

1. Make sure Python is installed.
2. Install required libraries:
    - numpy
    - matplotlib
    - seabreeze
3. Run the script:
4. python spectrometer_tool.py
5. If no spectrometer is detected, the tool automatically switches to dummy mode.

3. Interface Overview



The interface consists of:

- A live plot area showing spectra
- Sliders for integration time, smoothing, and ROI
- Checkboxes to show or hide curves
- Buttons for saving data and capturing references
- A status box displaying current settings

4. Typical Lab Workflow

1. Start the tool.
2. Block the light path and press **Dark** to record the dark reference.
3. Apply the reference condition and press **White**.
4. Adjust integration time and smoothing if needed.
5. Observe corrected transmission or absorbance curves.
6. Save data manually or enable auto-save.

5. Keyboard Shortcuts

Key	Action
d	Capture dark reference
w	Capture white reference
s	Save current spectrum
a	Toggle auto-save
p	Pause / resume acquisition
o	Choose output folder
q	Quit application

6. Output Data

Saved CSV files contain wavelength and intensity values.
Reference files and measurement data are stored in the selected output folder, allowing easy reuse and post-processing.

7. Notes for Students

- Always capture dark and white references before using corrected or absorbance views.
- Large spikes in corrected data usually indicate poor reference quality.
- Use smoothing and ROI to stabilize peak detection.